STACKIFICATION COMMUTES WITH 2-FIBRE PRODUCTS OF FIBRED CATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Universal property of stackification. lemma-stackify-universal-property (same chapter, named). Extends the canonical map from the fibre product to the fibre product of the stackifications, giving h : W -> X' x_{Y'} Z'.
- Stack criterion for a morphism. lemma-characterize-ff (same chapter, named). Reduces equivalence of h to isomorphisms of Mor sheaves plus local essential surjectivity.
- Essential surjectivity criterion. lemma-characterize-essentially-surjective-when-ff (same chapter, named). Supplies the local-essential-image condition paired with the Mor-sheaf isomorphisms.
- Construction of the stackification. lemma-stackify (same chapter, named). Identifies Mor sheaves of W locally with sheafified fibre products of Mor presheaves.

Source reference. Original lemma 026G: sources/stacks-stacks/pinned-original-source.tex, source lines 939–939. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 046N: sources/stacks-stacks/pinned-original-source.tex, source lines 1062–1062. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04WQ: sources/stacks-stacks/pinned-original-source.tex, source lines 1009–1009. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02ZN: sources/stacks-stacks/pinned-original-source.tex, source lines 1697–1697. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:stacks.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0435: sources/stacks-stacks/pinned-original-source.tex, source lines 1948–1948. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04TP: focused-seeds/geometry/S9-stk-stackification-fibre-product-fibred-categories/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $f : \mathcal{X} \to \mathcal{Y}$ and $g : \mathcal{Z} \to \mathcal{Y}$ be morphisms of fibred categories over $\mathcal{C}$. In this case the stackification of the 2-fibre product is the 2-fibre product of the stackifications.*

Proof. Let us denote $\mathcal{X}', \mathcal{Y}', \mathcal{Z}'$ the stackifications and $\mathcal{W}$ the stackification of $\mathcal{X} \times_{\mathcal{Y}} \mathcal{Z}$. By construction of 2-fibre products there is a canonical 1-morphism $\mathcal{X} \times_{\mathcal{Y}} \mathcal{Z} \to \mathcal{X}' \times_{\mathcal{Y}'} \mathcal{Z}'$. As the second 2-fibre product is a stack (see Lemma 026G) this 1-morphism induces a 1-morphism $h : \mathcal{W} \to \mathcal{X}' \times_{\mathcal{Y}'} \mathcal{Z}'$ by the universal property of stackification, see Lemma 0435. Now h is a morphism of stacks, and we may check that it is an equivalence using Lemmas 04WQ and 046N.

Thus we first prove that h induces isomorphisms of Mor-sheaves. Let ξ, ξ' be objects of $\mathcal{W}$ over $U \in \mathrm{Ob}(\mathcal{C})$. We want to show that

$$h : Mor(\xi, \xi') \longrightarrow Mor(h(\xi), h(\xi'))$$

is an isomorphism. To do this we may work locally on U (see Sites, Section 04TP). Hence by construction of $\mathcal{W}$ (see Lemma 02ZN) we may assume that ξ, ξ' actually come from objects (x, z, α) and (x', z', α') of $\mathcal{X} \times_{\mathcal{Y}} \mathcal{Z}$ over U. By the same lemma once more we see that in this case $Mor(\xi, \xi')$ is the sheafification of

$$V/U \longmapsto \mathrm{Mor}_{\mathcal{X}_V}(x|_V, x'|_V) \times_{\mathrm{Mor}_{\mathcal{Y}_V}(f(x)|_V, f(x')|_V)} \mathrm{Mor}_{\mathcal{Z}_V}(z|_V, z'|_V)$$

and that $Mor(h(\xi), h(\xi'))$ is equal to the fibre product

$$Mor(i(x), i(x')) \times_{Mor(j(f(x)), j(f(x'))} Mor(k(z), k(z'))$$

where $i : \mathcal{X} \to \mathcal{X}'$, $j : \mathcal{Y} \to \mathcal{Y}'$, and $k : \mathcal{Z} \to \mathcal{Z}'$ are the canonical functors. Thus the first displayed map of this paragraph is an isomorphism as sheafification is exact (and hence the sheafification of a fibre product of presheaves is the fibre product of the sheafifications).

Finally, we have to check that any object of $\mathcal{X}' \times_{\mathcal{Y}'} \mathcal{Z}'$ over U is locally on U in the essential image of h. Write such an object as a triple (x', z', α). Then x' locally comes from an object of $\mathcal{X}$, z' locally comes from an object of $\mathcal{Z}$, and having made suitable replacements for x', z' the morphism α of $\mathcal{Y}'_U$ locally comes from a morphism of $\mathcal{Y}$. In other words, we have shown that any object of $\mathcal{X}' \times_{\mathcal{Y}'} \mathcal{Z}'$ over U is locally on U in the essential image of $\mathcal{X} \times_{\mathcal{Y}} \mathcal{Z} \to \mathcal{X}' \times_{\mathcal{Y}'} \mathcal{Z}'$, hence a fortiori it is locally in the essential image of h. □